

Manikanta Gopi

Senior Backend & Platform Engineer

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SUMMARY

Backend and platform engineer with 4+ years of production experience building and operating backend systems, distributed workloads, cloud infrastructure and Kubernetes platforms. Experienced across Ruby on Rails, Go, TypeScript and Python, with AWS, GCP and Azure. Strong focus on infrastructure cost optimization, reliability, performance, cloud migration and asynchronous systems. Known for finding the real production constraint, making targeted changes, and verifying the result.

SELECTED ENGINEERING IMPACT

Video processing: Reduced video-processing infrastructure from roughly \$18,000/month to ~\$1,800/month by moving Lambda + FFmpeg workloads to in-house Kubernetes workers.

Database: Identified resource-intensive database queries and changed the surrounding systems, contributing roughly \$500/month in savings.

Kubernetes capacity: Introduced Karpenter with Spot + On-Demand capacity for resource-intensive workloads, reducing costs by roughly \$400/month.

S3 lifecycle: Added expiry for ephemeral media, saving roughly \$800 in May and preventing storage costs from growing unchecked.

Cloud migration: Led a production Azure AKS to AWS EKS migration using Terraform, service-by-service cutover, DNS/CDN migration and deliberate decommissioning with no maintenance window.

Queueing: Migrated queueing infrastructure from Redis to Dragonfly without downtime.

EXPERIENCE

Senior Software Engineer | Circleapp Online Services (Praja)

Apr 2025 - Present

Platform and infrastructure ownership alongside product engineering, including a GitOps-managed Kubernetes fleet, media processing, release infrastructure and production reliability.

- Own the GitOps repository describing ~24 production services on AWS EKS, including Helm charts, environment configuration and automated image updates.
- Reduced video-processing infrastructure from ~\$18k/month to ~\$1.8k/month by moving Lambda + FFmpeg workloads to in-house Kubernetes workers.
- Optimized database workloads and storage lifecycle, contributing ~\$500/month in DB savings and ~\$800 saved in May from S3 expiry policies.
- Enabled Karpenter with Spot + On-Demand capacity for resource-intensive workloads, reducing costs by roughly \$400/month.
- Migrated queueing infrastructure from Redis to Dragonfly without downtime.
- Moved background removal from a hosted vision API to a self-hosted ONNX GPU service in the Kubernetes cluster.

Software Engineer | Circleapp Online Services (Praja)

May 2022 - Mar 2025

Backend engineering across the Rails monolith and surrounding services, growing into distributed systems, infrastructure and cloud migration work.

- Built and operated a real-time messaging service using NestJS, Socket.IO, Redis pub/sub, MongoDB and BullMQ.
- Led production migration from Azure AKS to AWS EKS using Terraform, with service-by-service cutover and no maintenance window.
- Implemented subscription lifecycle workflows including charge scheduling, grace periods, refunds and idempotent payment handling.
- Built asynchronous poster and video generation workflows supporting the product's paid tier.
- Published sidekiq-assured-jobs, an open-source Ruby gem for recovering in-flight Sidekiq jobs after worker termination.

Software Engineer Intern | Circleapp Online Services (Praj)

Dec 2021 - Apr 2022

Started on the Flutter client and moved into backend engineering on the Rails API.

- Shipped chat features including conversation types, message deletion, link previews and member lists.
- Built and maintained the Rails API endpoints supporting those product features.

Machine Learning Intern | Continual Engine

2021

Applied machine learning internship focused on image understanding for accessibility tooling.

- Built image-to-text models in PyTorch, including autoencoders for feature extraction.
- Ran model training and evaluation workflows to measure improvements.

SELECTED PROJECTS

Seeker - Search Engine from First Principles - Built an inverted index, BM25, Levenshtein automata, FSTs, BKD trees and a byte-level segment format in ~20,000 lines of TypeScript; documented the work in a 43-chapter book.

Media Processing Platform - Built KEDA-scaled media workers driven by queue depth for poster and video generation, using NestJS, FFmpeg, Puppeteer, Redis and S3.

Real-time Messaging - Designed and operated cross-instance fan-out with Socket.IO and Redis pub/sub, MongoDB persistence and BullMQ for deferred work.

Self-hosted Vision Service - Deployed ONNX Runtime GPU inference in Kubernetes and validated a face-detector replacement against 200 real production images before shipping.

TECHNICA5 T30 Service - Deployed ONNEalows to measure improvements.